

Python: exercises on iteration

This notebook was generated in **pseudocode** mode.

Exercise: Print Numbers from 1 to N

Exercise Description

Write a function that prints all numbers from 1 to N using a for loop and range.

Your solution

```
def print_numbers(n):  
    # Step 1: Set up a for loop to iterate from 1 to n (inclusive)  
    # Step 2: For each number in the range, print it
```

Test your solution

```
# Test case 1: Test with small number  
print_numbers(5)
```

Debug your solution

The following cells contain code to generate links to Python Tutor for debugging your code when running the tests. **You need to first run the cell containing your solution code before running the cell containing the Python Tutor link.**

```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pyth
```

Exercise: Sum of First N Natural Numbers

Exercise Description

Write a function to calculate the sum of the first N natural numbers ($1 + 2 + 3 + \dots + N$) using a while loop.

Your solution

```
def sum_natural_numbers(n):  
    # Step 1: Initialize sum variable to 0 and counter to 1  
    # Step 2: Set up while loop that continues while counter <= n  
    # Step 3: Add current counter value to sum  
    # Step 4: Increment counter by 1  
    # Step 5: Return the final sum
```

Test your solution

```
# Test case 2: Test with another number
result = sum_natural_numbers(3)
expected = 1 + 2 + 3 # 6
assert result == expected, f"Expected {expected}, got {result}"
print(f"✓ sum_natural_numbers(3) = {result}")
```

Test your solution

```
# Test case: Example from solution
result = sum_natural_numbers(5)
print(f"Sum of first 5 natural numbers: {result}") # 15
```

Debug your solution

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```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pyth
```

Exercise: Print Even Numbers from 2 to 20

Exercise Description

Write a function that prints all even numbers between 2 and 20 using a for loop and range.

Your solution

```
def print_even_numbers():  
    # Step 1: Set up a for loop with start=2, stop=21, step=2  
    # Step 2: For each even number in the range, print it
```

Test your solution

```
# Test case 2: Verify range and step  
# Expected: prints 2, 4, 6, 8, 10, 12, 14, 16, 18, 20 (each on new line)  
print_even_numbers()
```

Debug your solution

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```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pyth
```

Exercise: Count Down from N to 1 (If Even)

Exercise Description

Write a function to print numbers from N to 1 using a while loop, but only if N is even.

Your solution

```
def countdown_if_even(n):  
    # Step 1: Check if n is even using modulo operator  
    # Step 2: If n is even, set up while loop that continues while n > 0  
    # Step 3: Print current n value  
    # Step 4: Decrement n by 1  
    # Step 5: If n is odd, do nothing
```

Test your solution

```
# Test case 2: Test with odd number (should print nothing)  
countdown_if_even(3)
```

Test your solution

```
# Test case 2 (assert)  
countdown_if_even(6) # Prints 6, 5, 4, 3, 2, 1
```

Debug your solution

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```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pyth
```

Exercise: Multiplication Table of 3

Exercise Description

Write a function that prints the multiplication table of 3 (from 3x1 to 3x10) using a for loop.

Your solution

```
def multiplication_table_3():  
    # Step 1: Set up a for loop to iterate from 1 to 10  
    # Step 2: For each multiplier, calculate 3 * multiplier  
    # Step 3: Print the multiplication equation and result
```

Test your solution

```
# Test case 2: Verify format  
# Expected: prints 3 x 1 = 3, 3 x 2 = 6, ..., 3 x 10 = 30  
multiplication_table_3()
```

Debug your solution

The following cells contain code to generate links to Python Tutor for debugging your code when running the tests. **You need to first run the cell containing your solution code before running the cell containing the Python Tutor link.**

```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pyth
```

Exercise: Sum of First N Even Numbers

Exercise Description

Write a function to calculate the sum of the first N even numbers (2 + 4 + 6 + ...) using a while loop.

Your solution

```
def sum_first_n_even(n):  
    # Step 1: Initialize sum to 0, current even number to 2, and count to 0  
    # Step 2: Set up while loop that continues while count < n  
    # Step 3: Add current even number to sum  
    # Step 4: Move to next even number (add 2)  
    # Step 5: Increment count  
    # Step 6: Return the final sum
```

Test your solution

```
# Test case 2: Test with another number  
result = sum_first_n_even(5)  
expected = 2 + 4 + 6 + 8 + 10 # 30  
assert result == expected, f"Expected {expected}, got {result}"  
print(f"✓ sum_first_n_even(5) = {result}")
```

Debug your solution

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```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pyth
```

Exercise: Numbers Divisible by 5 (1 to 50)

Exercise Description

Write a function that prints all numbers divisible by 5 between 1 and 50 using a for loop and range.

Your solution

```
def print_divisible_by_5():  
    # Step 1: Set up a for loop with start=5, stop=51, step=5  
    # Step 2: For each number divisible by 5, print it
```

Test your solution

```
# Test case 2: Verify range  
# Expected: prints 5, 10, 15, 20, 25, 30, 35, 40, 45, 50  
print_divisible_by_5()
```


Debug your solution

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```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pyth
```

Exercise: Factorial of 5

Exercise Description

Write a function to find the factorial of 5 using a while loop.

Your solution

```
def factorial_5():  
    # Step 1: Initialize factorial to 1 and counter to 5  
    # Step 2: Set up while loop that continues while counter > 0  
    # Step 3: Multiply factorial by current counter value  
    # Step 4: Decrement counter by 1  
    # Step 5: Return the final factorial
```

Test your solution

```
# Test case 2: Verify result
result = factorial_5()
assert result == 120, f"Expected 120, got {result}"
print(f"✓ Factorial of 5 is {result}")
```

Debug your solution

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```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pyth
```

Exercise: Sum of Odd Numbers (1 to 20)

Exercise Description

Write a function to calculate the sum of all odd numbers between 1 and 20 using a for loop.

Your solution

```
def sum_odd_numbers():
    # Step 1: Initialize sum to 0
    # Step 2: Set up for loop with start=1, stop=21, step=2 (odd numbers)
```

```
# Step 3: Add each odd number to sum  
# Step 4: Return the final sum
```

Test your solution

```
# Test case 2: Verify result  
result = sum_odd_numbers()  
assert result == 100, f"Expected 100, got {result}"  
print(f"✓ Sum of odd numbers 1-20 is {result}")
```

Test your solution

```
# Test case: Example from solution  
result = sum_odd_numbers()  
print(f"Sum of odd numbers 1-20: {result}") # 100
```

Debug your solution

The following cells contain code to generate links to Python Tutor for debugging your code when running the tests. **You need to first run the cell containing your solution code before running the cell containing the Python Tutor link.**

```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pyth
```

Exercise: Numbers Divisible by K (1 to N)

Exercise Description

Write a function to print numbers between 1 and N that are divisible by K using a for loop and range.

Your solution

```
def print_divisible_by_k(n, k):  
    # Step 1: Set up for loop with start=k, stop=n+1, step=k  
    # Step 2: For each number divisible by k, print it
```

Test your solution

```
# Test case 2: Test with n=20, k=4  
# Expected: prints 4, 8, 12, 16, 20  
print_divisible_by_k(20, 4)
```

Test your solution

```
# Test case: Example from solution  
print_divisible_by_k(20, 3) # Prints 3, 6, 9, 12, 15, 18
```

Debug your solution

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```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pyth
```

Exercise: Count Digits in a Number

Exercise Description

Write a function that counts how many digits are in a positive integer.

Your solution

```
def count_digits(num):  
    # Step 1: Initialize a counter to 0  
    # Step 2: While num > 0, do integer division by 10 and increment counter  
    # Step 3: Return the counter
```

Test your solution

```
assert count_digits(123456) == 6  
assert count_digits(7) == 1  
assert count_digits(10) == 2  
print("✓ count_digits tests passed")
```

Debug your solution

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```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pyth
```

Exercise: Cubes from 1 to 5

Exercise Description

Write a function that returns a list with the cubes of numbers from 1 to 5 using a for loop.

Your solution

```
def cubes_1_to_5():  
    # Step 1: Initialize an empty list  
    # Step 2: For i from 1 to 5 inclusive, append i**3 to the list  
    # Step 3: Return the list
```

Test your solution

```
assert cubes_1_to_5() == [1, 8, 27, 64, 125]  
print("✓ cubes_1_to_5 tests passed")
```

Debug your solution

The following cells contain code to generate links to Python Tutor for debugging your code when running the tests. **You need to first run the cell containing your solution code before running the cell containing the Python Tutor link.**

```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pyth
```

Exercise: Product of Numbers from 1 to N

Exercise Description

Write a function that returns the product of numbers from 1 to N using a for loop.

Your solution

```
def product_1_to_n(n):  
    # Step 1: Initialize product to 1  
    # Step 2: For i from 1 to n inclusive, multiply product by i  
    # Step 3: Return product
```

Test your solution

```
assert product_1_to_n(1) == 1  
assert product_1_to_n(3) == 6  
assert product_1_to_n(5) == 120  
print("✓ product_1_to_n tests passed")
```

Debug your solution

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```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pyth
```

Exercise: Odd Numbers from 1 to 15

Exercise Description

Write a function that returns all odd numbers from 1 to 15 using a while loop.

Your solution

```
def odd_1_to_15():  
    # Step 1: Initialize i to 1 and an empty list  
    # Step 2: While i <= 15, append i then increment by 2  
    # Step 3: Return the list
```

Test your solution

```
assert odd_1_to_15() == [1, 3, 5, 7, 9, 11, 13, 15]  
print("✓ odd_1_to_15 tests passed")
```


Debug your solution

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```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pyth
```

Exercise: Print Numbers from 5 to 1

Exercise Description

Write a function that returns numbers from 5 to 1 using a for loop with range.

Your solution

```
def countdown_5_to_1():  
    # Step 1: Initialize an empty list  
    # Step 2: Use range from 5 down to 1 inclusive and append to list  
    # Step 3: Return the list
```

Test your solution

```
assert countdown_5_to_1() == [5, 4, 3, 2, 1]  
print("✓ countdown_5_to_1 tests passed")
```

Debug your solution

The following cells contain code to generate links to Python Tutor for debugging your code when running the tests. **You need to first run the cell containing your solution code before running the cell containing the Python Tutor link.**

```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pyth
```

Exercise: Sum of the First 7 Odd Numbers

Exercise Description

Write a function to find the sum of the first 7 odd numbers using a while loop.

Your solution

```
def sum_first_7_odds():  
    # Step 1: Initialize i=1, count=0, total=0  
    # Step 2: While count < 7, add i to total, increment i by 2 and count by 1  
    # Step 3: Return total
```

Test your solution

```
assert sum_first_7_odds() == 49  
print("✓ sum_first_7_odds tests passed")
```

Debug your solution

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```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pyth
```

Exercise: Repeat Until "stop"

Exercise Description

Write a function that, given a sequence of strings, counts how many entries occur until the string "stop" appears.

Your solution

```
def until_stop_count(inputs):  
    # Step 1: Initialize a counter to 0  
    # Step 2: For each string in inputs:  
    #     If the string is "stop", break the loop  
    #     Otherwise increment the counter  
    # Step 3: Return the counter
```

Test your solution

```
assert until_stop_count(["a", "b", "stop", "x"]) == 2
assert until_stop_count(["stop"]) == 0
assert until_stop_count(["hello", "world"]) == 2
print("✓ until_stop_count tests passed")
```

Debug your solution

The following cells contain code to generate links to Python Tutor for debugging your code when running the tests. **You need to first run the cell containing your solution code before running the cell containing the Python Tutor link.**

```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pyth
```

Exercise: Repeat String as Many Times as Its Length

Exercise Description

Write a function that takes a string and returns a list with the string repeated as many times as its length.

Your solution

```
def repeat_string_len_times(s):
    # Step 1: Initialize an empty list
```

```
# Step 2: For i in range(len(s)), append s
# Step 3: Return the list
```

Test your solution

```
assert repeat_string_len_times("abc") == ["abc", "abc", "abc"]
assert repeat_string_len_times("") == []
print("✓ repeat_string_len_times tests passed")
```

Debug your solution

The following cells contain code to generate links to Python Tutor for debugging your code when running the tests. **You need to first run the cell containing your solution code before running the cell containing the Python Tutor link.**

```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pyth
```

Exercise: Ask for a Password

Exercise Description

Write a function that, given a sequence of password attempts, returns the number of attempts needed to enter the correct password ("password123"). Return -1 if never entered.

Your solution

```
def password_attempts(attempts, correct="password123"):
    # Step 1: Initialize attempts counter to 0
    # Step 2: For each attempt in attempts:
    #     Increment attempts counter
    #     If attempt equals correct, return the counter
    # Step 3: If loop finishes without match, return -1
```

Test your solution

```
assert password_attempts(["x", "y", "password123"]) == 3
assert password_attempts(["password123"]) == 1
assert password_attempts(["a", "b"]) == -1
print("✓ password_attempts tests passed")
```

Debug your solution

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```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pyth
```

